

Grid-Connected Mode

87L Line Differential (TR-37)
 $k_{ibr} \geq 0.06$, 20 ms

Z1 Quadrilateral (TR-34)
20 ms, $k \geq 0.10$

67Q/67N Directional (TR-38)
30 ms, GFM: $k \geq 0.10$

Z2/POTT (TR-33)
32 ms, $k \geq 0.10$

OC Group 1: $I_p = 1.50$ pu (TR-40)
backup, 706 ms

87B Bus Diff. (TR-41)
20 ms, busbar only

Islanded Mode

87L DISABLED
(comms severed, TR-39)

Z1/Z2 DISABLED
(no remote infeed)

67/OC Directional (TR-40)
 $k \geq 0.0075$

POTT DISABLED
(no carrier path)

OC Group 2: $I_p = 0.30$ pu (TR-40)
primary, 413–527 ms

87B Bus Diff. (TR-41)
20 ms (always active)

GOOSE
120 ms
↔

Mode switch triggered by TR-39 ROCOF/V/f detection
Setting groups 1↔2 changed via IEC 61850 GOOSE
87B remains active in both modes (continuously)